

Communication and data transmission MID exam .

1) The interface circuit between your personal computer & wired telephone network is :

- a. Antenna
- b. Codec
- c. Modem
- d. Fiber

2) DSL stands for

- a. Digital single line
- b. Digital system line
- c. Digital submission line
- d. Digital subscriber line

3) the most difficult medium to tape is :

- a. Wireless medium
- b. Fiber
- c. Coax cable
- d. Twisted pair

4) given a channel with capacity of 20 Mbps & Bandwidth of 3MHz . Assuming white noise , what SNR is required to achieve the capacity ?

- a. 101 dB
- b. 102 dB
- c. 100 dB
- d. 103 dB

5)if an amplifier has 30 dB voltage gain . what voltage ratio does the gain represent

- a. 31.6 V
- b. 31.6
- c. 31.6 mV
- d. 3.16

6) The power line voltage in Jordan can be expressed as:

- a. $230 \sin(2\pi \cdot 50t)$
- b. $170 \sin(2\pi \cdot 60t)$
- c. $230 \sin(2\pi \cdot 60t)$
- d. $230 \sin(2\pi \cdot 360t)$

7) The most of energy in the signal is contained in :

- a. Spectrum
- b. Last harmonic
- c. Bandwidth
- d. Harmonics

8) Radio signals are made up to

- a. Voltages and currents
- b. Electrons and protons
- c. Noise and data
- d. Electric and magnetic fields

9) The communications medium causes the signal to be :

- a. Interfered with
- b. Attenuated
- c. Amplified
- d. Modulated

10) if signal has DC component , then :

- a. It has $f=0$ & a non-zero average amplitude
- b. It has $f \neq 0$ & a non-zero average amplitude
- c. It has $f \neq 0$ & a zero average amplitude
- d. It has $f=0$ & a zero average amplitude

11) Which of the following is correct:

- a. The higher data rate of signal , the greater is the bandwidth & the greater the distortion
- b. The higher data rate of signal , the greater is the bandwidth & the smaller the distortion
- c. The higher data rate of signal , the smaller is the bandwidth & the greater the distortion
- d. The smaller data rate of signal, the greater is the bandwidth & the greater the distortion

12) delay distortion is caused by :

- a. Different velocity of propagation for different harmonics
- b. Constant velocity of propagation for different harmonics
- c. Different velocity of propagation in free space for different harmonics
- d. Constant velocity of propagation in free space for different harmonics

13) the following medium is limited in distance , bandwidth & data rate

- a. Coax cable
- b. Waveguide
- c. Twisted pair
- d. Free space

14) the following mode is used in telephony

- a. Step index multimode
- b. Graded index multimode
- c. Single index multimode
- d. Single mode propagation

15) usually human vision allocated with respect to wavelength in

- a. Megameters
- b. Nanometers
- c. Millimeters
- d. Micrometers

16) Radio waves travel more slowly near the ground , because :

- a. The refractive index of atmosphere decreases with height
- b. The refractive index of atmosphere increases with height
- c. The refractive index of atmosphere the same regardless to the height
- d. The reflection index of atmosphere decreases with height

17) The combination of Tx/Rx in the satellite is called :

- a. Uplink
- b. Transponder
- c. Downlink
- d. Repeater

18) multipath could be :

- a. Could be advantage and could be disadvantage
- b. Always advantage
- c. Always disadvantage
- d. Always a merit

19) international audio broadcasting is usually used :

- a. Ground wave propagation
- b. Direct wave propagation
- c. Sky wave propagation
- d. Surface wave propagation

20) typical speech has a dynamic range of about :

- a. 2.5 dB
- b. 250 dB
- c. 0.25 Db
- d. 25 dB

Answers:-

1	2	3	4	5	6	7	8	9	10
C	D	B	A	B	A	C	D	B	A
11	12	13	14	15	16	17	18	19	20
B	A	C	D	B	A	B	A	C	D